Patch-Based Tokenisation of Discrete-Time Signals

Coursework — Points 1 & 2

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1 Point 1

1.1 Setup and Formal Definition

Let $x[n]$ be a discrete-time signal with $n \in \mathbb{Z}$, sampled at frequency f_s. Patch-based tokenisation extracts a sequence of tokens by sliding a window of size P samples with stride S:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \ldots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \qquad k = 0, 1, 2, \ldots \tag{1}$$

Definition 1 (Patch Tokenisation Operator)**.** *The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$ defined by (1) is called the* patch tokenisation operator*. It is linear: collecting K patches yields*

$$\mathbf{P} = \mathcal{T}\{x\} = \mathbf{W} \cdot \mathbf{x}, \tag{2}$$

where $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a binary selection matrix whose k-th block of P rows picks the P consecutive samples starting at index kS.

1.2 Condition for Two Consecutive Patches to Be Identical

Proposition 1. *Patches $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$ are identical if and only if*

$$x[n] = x[n+S] \quad \forall\, n \in \{kS,\, kS+1,\, \ldots,\, kS+P-1\}. \tag{3}$$

Globally, if the signal is periodic with period T_0 (in samples), identical consecutive patches occur whenever

$$\boxed{S = m \cdot T_0, \qquad m \in \mathbb{Z}^+.} \tag{4}$$

Equivalently, in terms of signal frequency f and sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \qquad m \in \mathbb{Z}^+. \tag{5}$$

Proof. $\mathbf{p}_k = \mathbf{p}_{k+1}$ component-wise requires $x[kS+j] = x[(k+1)S+j]$ for all $j \in \{0, \ldots, P-1\}$, i.e. $x[n] = x[n+S]$ on that window. For a periodic signal with $T_0 = f_s/f$ samples, this holds whenever S is an integer multiple of T_0, giving (4). $\square$

1.3 Loss of Observability

When $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all k, the token sequence is constant — the model sees the same vector repeated. The *rank* of the token matrix $\mathbf{P}$ collapses to 1. Consequently:

- Temporal variation is invisible to the model.
- Any two signals differing only by a phase shift of S samples produce the same token sequence: they are **observationally equivalent**.
- Even when patches are not fully identical but merely highly correlated ($S \approx mT_0$), the effective information in the token sequence is reduced (*soft* loss of observability).

2 Point 2

2.1 The Patch-Induced Frequency

The tokeniser introduces a characteristic *patch frequency*:

$$f_{\text{patch}} = \frac{f_s}{S}. \tag{6}$$

This is the rate at which new patches are produced (in Hz, referred to original time), analogous to a resampling rate in the token domain.

2.2 Phase-Locking Condition

Consider a sinusoidal signal $x[n] = A\cos\bigl(2\pi f n/f_s + \phi\bigr)$.

Definition 2 (Phase-Locking)**.** *The signal* phase-locks *to the patch grid when its period $T_0 = f_s/f$ (in samples) is* commensurate *with S:*

$$\frac{S \cdot f}{f_s} \in \mathbb{Q}. \tag{7}$$

The degenerate (critical) case occurs when this ratio is an integer:

$$f = m \cdot \frac{f_s}{S} = m \cdot f_{\text{patch}}, \qquad m \in \mathbb{Z}^+, \tag{8}$$

*i.e. f is a **harmonic of** f_{patch}.*

2.3 Effect of Patch Size P

A single patch of P samples spans a duration P/f_s seconds and covers $P \cdot f/f_s$ cycles of the signal. When

$$\frac{P \cdot f}{f_s} \in \mathbb{Z}^+, \quad \text{i.e.} \quad f = k \cdot \frac{f_s}{P}, \quad k \in \mathbb{Z}^+, \tag{9}$$

the patch contains an exact integer number of cycles: its content is *internally redundant*.

The most severe degeneracy occurs when both the stride and the patch-size conditions hold simultaneously:

$$f = \frac{\ell \cdot f_s}{\operatorname{lcm}(P, S)}, \qquad \ell \in \mathbb{Z}^+. \tag{10}$$

2.4 Token-Space Degeneracies

At $f = m\, f_s/S$ (harmonics of f_{patch}).

- All patches are translates of the same waveform by a phase that is a multiple of 2π: they become **identical** (cf. Point 1).
- The token sequence is constant; it encodes **no phase information**.
- Two signals at the same frequency but different phases (shifted by any multiple of S samples) map to the same token sequence: *token-space invariance*.

At $f = k\, f_s/P$ (but not harmonics of f_{patch}).

- Each patch is internally periodic, so its content is *within-patch redundant.*
- Inter-patch variation still exists (partial degeneracy).

2.5 Overlap, Stride, and Blind-Spot Frequencies

When $S < P$ (overlapping patches), consecutive patches share $P - S$ samples, partially mitigating the aliasing effect even near harmonic frequencies. For $S \geq P$ (no overlap or gaps), the aliasing is maximal; the **blind-spot frequencies** are:

$$\boxed{f_{\text{blind}} = m \cdot \frac{f_s}{S}, \qquad m = 1, 2, 3, \ldots, \left\lfloor \frac{S}{2} \right\rfloor.} \tag{11}$$

The upper limit follows from the Nyquist criterion applied to the token sequence, whose effective sample rate is f_s/S. This consegunces followed what expressed in [?]

2.6 Summary

Condition	Effect
$f = m\, f_s/S$	Full phase-locking → identical patches → constant token sequence
$f = k\, f_s/P$	Internal patch periodicity → within-patch redundancy
Both simultaneously	Maximal degeneracy
$S < P$ (overlap)	Partial mitigation; residual phase information
$S \geq P$ (no overlap)	Full degeneracy at blind-spot frequencies

Table 1: Summary of conditions and their effects on token-space observability.

References